

A counterexample to conjecture 00000008438

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Claim being disproved. The conjecture bounds the number of columns of a strength- t orthogonal array over n symbols by $n^2 - n + t$. Here “strength t ” means that on any t distinct rows every ordered t -tuple of symbols occurs equally often. We use an array of index one, so the example does not rely on repeating columns.

Construction. Take $n = 2$, $t = 3$, and the array

$$A = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}.$$

Its columns are all eight distinct binary words of length three. Hence any ordering of its three rows contains every ordered binary triple exactly once: reordering coordinates is a bijection on $\{0, 1\}^3$. Thus A is a strength-three, index-one orthogonal array.

Contradiction. The array has 8 columns, but the proposed bound is

$$n^2 - n + t = 2^2 - 2 + 3 = 5 < 8.$$

A counterexample to the first bound disproves the conjunction in the conjecture; the prime-power attainability and deficit assertions need not be considered.

A second witness with matching row and alphabet counts. Take instead the three-row array whose columns are all 27 words in $\{0, 1, 2\}^3$. It also has strength three and index one, since permuting its rows permutes all ternary triples. Here both the alphabet size and the row count equal 3, while $27 > 3^2 - 3 + 3 = 9$. Thus this example also avoids any possible ambiguity between those two meanings of n . The accompanying Lean theorems `ternaryArray_is_OA3`, `ternaryArray_columns_distinct`, and `conjecture8438_false_ternary` verify the full second witness and the resulting negation.

Formal verification and semantic correspondence. The accompanying `lean/Main.lean` uses only Lean 4.19.0 and `Std`. The type `Fin r` indexes rows, `Fin c` indexes columns, and `Fin n` indexes symbols. The definition `occurrences` counts actual columns matching prescribed entries. The predicate `OA3` quantifies over every triple of distinct rows, in every order, and every prescribed symbol triple. The theorem `binaryArray_is_OA3` verifies this property for the displayed array; `binaryArray_columns_distinct` also verifies that no two columns coincide.

`ColumnBoundAtThree` states the conjectured universal upper bound specialized to $t = 3$, over arbitrary alphabet size, row count, column count, and array. The final theorem `conjecture8438_false` proves its negation by instantiating that universal claim with A . Thus the formal result includes the orthogonal-array hypotheses and the implication to the claimed bound, rather than only the numerical inequality $8 > 5$.

All enumeration is checked by Lean’s kernel using `decide`; there is no external computation oracle. The printed axiom lists contain only the standard Lean axiom `propext`. No `sorry`, `admit`, `native_decide`, or additional axiom is used.

Reproduction. Run `lean Main.lean` with Lean 4.19.0. Successful compilation prints the axiom lists of the array property, column distinctness, and final disproof.